

RAHUL MITRA

SOFTWARE ENGINEER PROFILE

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PROFILE SUMMARY

Early-career graduate software engineer with NUS ISE, Computer Science, and Mathematics training, building production-minded systems across Python, React, TypeScript, cloud deployment, AI-assisted development, and open-source tooling.

EDUCATION

National University of Singapore

Aug 2023 - Jul 2027

- Bachelor of Engineering, Industrial Systems Engineering; Second Major in Computer Science; Minor in Mathematics
- Relevant coursework: Software Engineering; 3D Computer Vision; Artificial Intelligence; Data Structures and Algorithms; Simulation; Statistics; Operations Research

Anderson Serangoon Junior College

Feb 2019 - Dec 2020

- Singapore-Cambridge GCE A-Level

WORK AND INTERSHIP EXPERIENCE

Data Analytics and Product Management Intern, Abbott Laboratories Singapore Pte Ltd

Jan 2026 - Present

- Built a Python digital twin simulation and constraint-programming optimizer for an NP-hard hybrid flow shop scheduling problem, translating production complexity into decision-ready schedules.
- Architected an Azure-based web simulator for Advanced Process Control systems across global manufacturing plants, using Docker containerization and Bitbucket CI/CD for repeatable releases.
- Offered a follow-on 7-month Data Engineer, Operational AI Systems contract, extending work across manufacturing analytics, deployment reliability, and operational AI workflows.

Software Engineer (Citizen Developer), People's Association

Sep 2025 - Present

- Awarded the S\$20,000 Sparks Community Innovation Fund at the 2025 Community Hackathon to continue Churp for Tengah community gardening operations.
- Built product workflows for transparent plot allocation, resident feedback, and civic service delivery while conducting user testing to refine product requirements.

Base Support Assistant, Republic of Singapore Navy

Jan 2022 - Jan 2023

- Engineered an Excel VBA automation workflow that compressed a six-month food-waste analysis cycle into a sub-three-minute reporting process inside a security-constrained environment.
- Managed sensitive operational data under SAF confidentiality expectations while improving reliability, traceability, and analyst capacity.

SELECTED PROJECTS AND COMPETITIONS

OnTheSpectrum - Codex and Blender MCP 3D Pipeline

May 2026

- Built an innovative local-first Blender-to-Three.js asset pipeline using Codex, Blender MCP, generated GLB assets, metadata, previews, playable world QA, and Remotion video creation workflows.

AsyncDDGS - Open-Source Search Client

2024 - Present

- Maintained an asyncio-first, aiohttp-based DuckDuckGo search client on PyPI with pytest CI/CD and adoption across AI tooling, Discord bots, and search-enabled projects.

Churp - Sparks by PA Community Innovation

2025 - Present

- Built a citizen-developer platform for Tengah gardening plot transparency, allocation workflows, user feedback, and Smart Nation community operations.

Interactive Portfolio Infrastructure

2026

- Built a React, TypeScript, Three.js, Matter.js, and AI-assisted portfolio with profile switching, searchable field notes, analytics, and a 3D project world.

SmartExam - GovTech x NTUPC Product Hackathon

2025

- Built a multi-agent autonomous exam generation system with RAG, Next.js, source ingestion, question generation, and education workflow automation.

TECHNICAL SKILLS AND CERTIFICATIONS

Skills: Python; TypeScript; React; Next.js; Three.js; FastAPI; Flask; aiohttp; asyncio; Java; Docker; CI/CD; Azure; Codex; Blender MCP; Remotion.

- NVIDIA Disaster Risk Monitoring Using Satellite Imagery - geospatial AI, U-Net, satellite data, GPU workflows.
- Packt Deep Reinforcement Learning Hands-On - DQN, Double DQN, DDPG, PPO, A2C, Python implementation.

CO-CURRICULAR AND COMMUNITY

Elderly Care Volunteer, NTUC Health Active Ageing Centre

Sep 2025 - Present

- Supported senior engagement activities, strengthening empathy for healthcare, accessibility, and community product design.

Astronomy Head and Robotics Member, STEM Inc. (ASRJC)

2019 - 2020

- Led astronomy outreach, competed in the Singapore Astronomical Olympiad, and built autonomous robotics systems for national competitions.